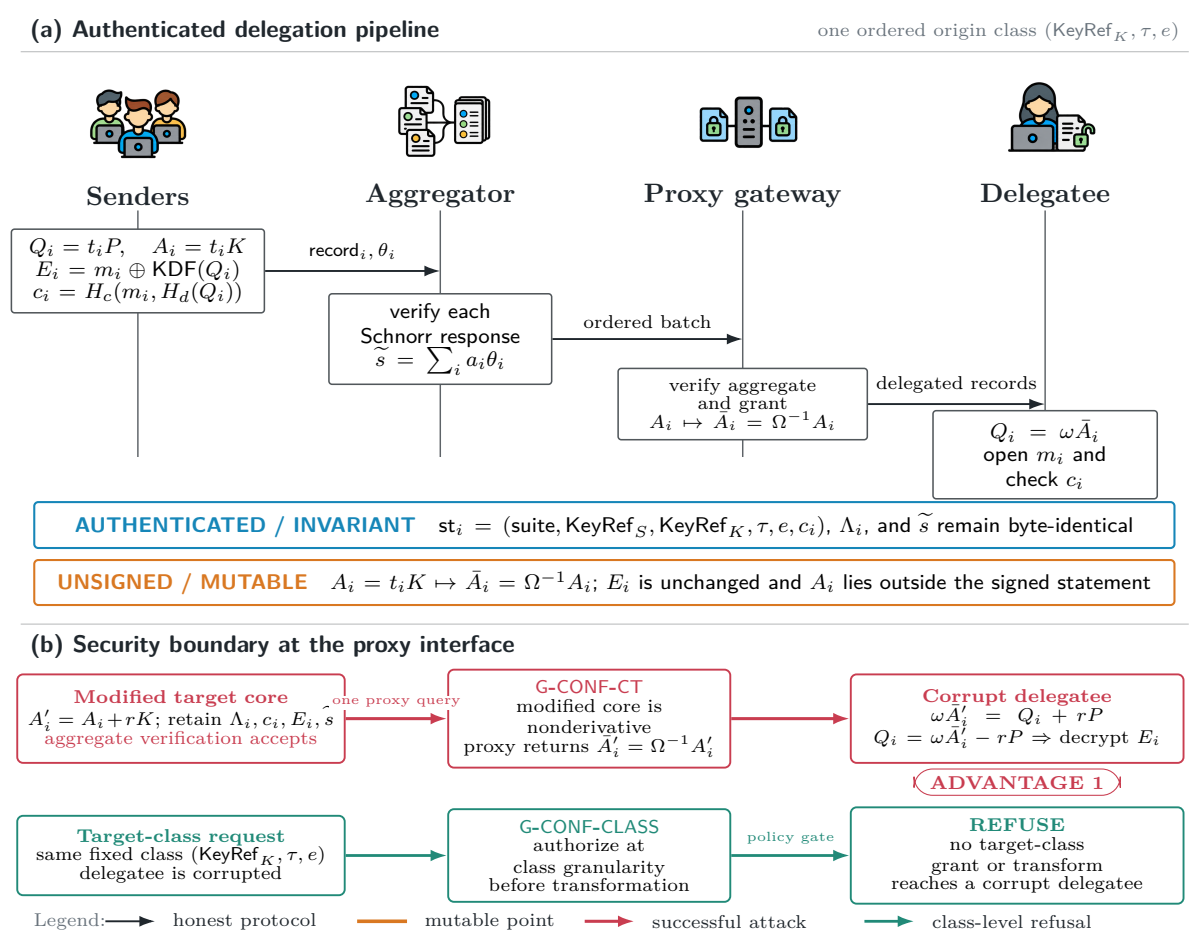

(a) Authenticated delegation pipeline
one ordered origin class (KeyRef$_K$, τ, e)
Senders
Aggregator
Proxy gateway
Delegatee
$Q_i = t_iP, \quad A_i = t_iK$
$E_i = m_i \oplus \mathsf{KDF}(Q_i)$
$c_i = H_c(m_i, H_d(Q_i))$
record$_i$, θ_i
verify each Schnorr response
$\widetilde{s} = \sum_i a_i\theta_i$
ordered batch
verify aggregate and grant
$A_i \mapsto \bar{A}_i = \Omega^{-1}A_i$
delegated records
$Q_i = \omega\bar{A}_i$
open m_i and check c_i
AUTHENTICATED / INVARIANT
st$_i$ = (suite, KeyRef$_S$, KeyRef$_K$, τ, e, c_i), Λ_i, and $\widetilde{s}$ remain byte-identical
UNSIGNED / MUTABLE
$A_i = t_iK \mapsto \bar{A}_i = \Omega^{-1}A_i$; E_i is unchanged and A_i lies outside the signed statement
(b) Security boundary at the proxy interface
Modified target core
$A'_i = A_i + rK$; retain $\Lambda_i, c_i, E_i, \widetilde{s}$
aggregate verification accepts
one proxy query
G-CONF-CT
modified core is nonderivative
proxy returns $\bar{A}'_i = \Omega^{-1}A'_i$
Corrupt delegatee
$\omega\bar{A}'_i = Q_i + rP$
$Q_i = \omega\bar{A}'_i - rP \Rightarrow$ decrypt E_i
ADVANTAGE 1
Target-class request
same fixed class (KeyRef$_K$, τ, e)
delegatee is corrupted
G-CONF-CLASS
authorize at class granularity
before transformation
policy gate
REFUSE
no target-class grant or transform reaches a corrupt delegatee
Legend:
honest protocol
mutable point
successful attack
class-level refusal